

June 22, 2026

Dr. Kevin Groot Lipman
fomo.fo lab
Netherlands Cancer Institute

Dear Dr. Kevin Groot Lipman and the Hiring Committee,

I am writing to express my strong interest in the PhD Student position focusing on Foundation Models for Oncology at the Netherlands Cancer Institute (NKI) and the University of Amsterdam.

As an AI & Data Science Engineer specializing in medical deep learning, I am highly motivated by the challenge of applying large-scale foundation models to clinical oncology. During my "Cardiovascular Medical Digital Twin" Master's thesis, I built architectures capable of handling complex, real-world clinical data. Specifically, I engineered a "Missingness-Aware Gate" to dynamically handle sensor dropouts and sparse clinical records without corrupting the latent state of the model. I also engineered Medical_MambaTS, a linear-time state-space model that ensures mathematical stability during long-context medical inference.

I believe my experience in robust, high-frequency medical modeling and handling missing clinical data perfectly aligns with the complexities of multi-modal oncology foundation models. The clinical focus of the fomo.fo lab—specifically the goal of moving beyond narrow task-specific models to general-purpose patient representations—is the exact research trajectory I wish to pursue for my doctorate.

Please find attached my complete application package, including my research statement and technical portfolio.

Thank you for your time and consideration. I would welcome the opportunity to discuss my qualifications with you in an interview.

Sincerely,
Ismail Aissaoui
<https://ismail-aissaoui.vercel.app>

Research Statement: Foundation Models for Oncology (fomo.fo)

Initial Research Ideas and Starting Point

The transition from task-specific oncology models to general-purpose Foundation Models requires overcoming the inherent “messiness” of clinical data: specifically, missing modalities (e.g., a patient has an MRI but no histopathology) and irregular temporal sampling (e.g., uneven intervals between patient check-ups).

To start this research, I propose focusing initially on the **Temporal and Multimodal Alignment Problem**. Rather than immediately attempting to pre-train a massive 3D volumetric model from scratch, I would propose a modular, state-space approach:

1. **Missingness-Aware Tokenization:** Building on the “Missingness-Aware Gate” I developed for my cardiovascular digital twin, I propose creating a tokenizer that explicitly embeds the “absence” of a modality as a learnable token rather than a zero-padded vector. This allows the foundation model to reason about *why* a test was not ordered, which often contains clinical signal.
2. **Continuous-Time State Space Modeling (SSM):** To handle irregular longitudinal patient data (e.g., tracking disease progression over months or years), standard Transformers struggle due to fixed positional encodings. I propose adapting linear-time SSMs (like Mamba) to continuous-time differential equations. This allows the model to interpolate patient state spaces at any arbitrary time t , effectively modeling the continuous progression of a tumor even when discrete scans are sparse.

This initial approach leverages my existing expertise in medical time-series and state-space models, acting as a stepping stone toward full 3D multimodal integration.

AISSAOUI Ismail

AI & Data Science Engineer

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PROFILE

AI & Data Science Engineer with a background in Full-Stack Software Engineering. Experienced in building projects across diverse domains, including Industrial Digital Twins, Healthcare, Financial Trading, Computer Vision, and Enterprise Management Systems. My primary interests include Deep Learning, Graph Neural Networks, Reinforcement Learning, and Natural Language Processing (NLP).

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL

ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, Lightning, ONNX

AI Techniques: Transformers (BERT, GPT, RAG), GNN, GAN, XGBoost, LSTM

Optimization: PSO, Genetic Algorithm, ACO, Simulated Annealing, Bayesian Optimization

Web & API: React, Next.js, FastAPI, Node.js, TailwindCSS

Data & MLOps: MLflow, Airflow, Kubeflow, DVC, Pandas, NumPy

Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra

Cloud & DevOps: Docker, Linux, Git, CI/CD, CUDA, AWS, Grafana

PROFESSIONAL EXPERIENCE

Graduate AI Research Engineer (Intern)

Jan 2026 – Jul 2026

Sonatrach (Industrial Surveillance Division) — Algeria

Architected an AI-native Digital Twin for real-time monitoring of Gas Turbines.

- Architected a 3-tier Dual-Track framework decoupling safety-critical Edge models from Cloud analytics.
- Developed **TCNEdge** achieving **0.049ms INT8 ONNX C++ inference** for emergency shutdown.
- Engineered **MambaTS** with a missingness-aware gate to maintain robustness during sensor dropouts.
- Implemented **KAN-PCNN** and **ST-KDFormer** to ensure strict thermodynamic consistency at scale.

Network Engineering Intern

Sep 2024

Algérie Télécom RMC Center — Chlef, Algeria

EDUCATION

National Higher School of Computer Science (ESI-SBA)

2021–2026

Engineering and Master's Degree in Computer Science

Specialization: Artificial Intelligence & Data Science

CERTIFICATIONS

DeepLearning.AI: Machine Learning Specialization, Deep Learning Specialization, NLP Specialization

ZTM Academy: PyTorch: Zero to Mastery, TensorFlow Developer

ACADEMIC THESES

State Engineering Degree Thesis:

Physics-Informed Explainable Digital Twin for Industrial Gas Turbines: A Dual-Track Cloud-Edge Deep Learning Framework with Koopman-Augmented State-Space Models

Master's Degree Thesis:

Hemodynamic Digital Twins: Fusing Hemodynamic Physics-Informed Neural Networks and Mamba-SSM for ICU Patient Prognostics

PROJECTS

Industrial Gas Turbine Digital Twin (Sonatrach)

2026

- **Edge & Cloud Architecture:** Engineered a 3-tier framework decoupling safety-critical TCNEdge (Siemens PLC) from near-edge MambaTS and cloud-based ST-KDFormer.
- **Ultra-Low Latency Inference:** Compiled 1D Dilated Convolutions (TCNEdge) to INT8 ONNX C++, achieving 0.049ms latency for emergency shutdown.
- **Physcis & Robustness:** Implemented MambaTS with missingness-aware gating for broken sensors, and KAN-PCNN with PyTorch Autograd for absolute thermodynamic consistency.

Tech: MambaTS, TCNEdge, ST-KDFormer, KAN-PCNN, PyTorch, C++ ONNX, PINN

Cardiovascular Medical Digital Twin

2026

- **High-Frequency Edge-Cloud Architecture:** Engineered **Medical_MambaTS** for sub-millisecond bedside inference and dual-branch **SST_KODA_MultiScale** for robust cloud forecasting using 100Hz VitalDB ICU telemetry.
- **Physics-Informed DeepONet:** Implemented a 0D Windkessel **PI_DeepONet** with PyTorch Autograd, forcing the model to obey cardiovascular fluid dynamics without retraining for new patients.
- **Data Pipeline & Robustness:** Developed a PyArrow Parquet zero-copy pipeline avoiding GPU starvation, and designed a custom missingness-aware gate to handle sensor dropouts natively.

Tech: PyTorch, Medical_MambaTS, PI_DeepONet, SST_KODA_MultiScale, Parquet PyArrow, Autograd

Aura — Enterprise AI Trading Platform

2025

- **Hybrid Forecasting:** Built an autonomous trading system combining LSTM networks for time-series predictions.
- **Strategic Decision-Making:** Implemented Deep Q-Learning (Reinforcement Learning) to optimize long-term reward policies.
- **Real-Time Dashboard:** Engineered a high-throughput FastAPI backend with WebSockets and a Next.js interactive UI.

Tech: TensorFlow, PyTorch, FastAPI, Next.js, WebSockets, RL

GNN-based Recruiting System (Career Connect)

2024–2025

- **Relational Learning:** Developed a recommendation engine using Graph Neural Networks (GNN) to map candidate-job relationships.
- **Semantic Embeddings:** Utilized PyTorch Geometric and SBERT for deep semantic text processing.
- **System Architecture:** Deployed via FastAPI and PostgreSQL, significantly outperforming keyword-based heuristics.

Tech: PyTorch Geometric, SBERT, FastAPI, PostgreSQL, GNN

LANGUAGES

Arabic: Native | English: Advanced | French: Professional

Applicant Profile & Supplementary Information

Ismail Aissaoui
<https://ismail-aissaoui.vercel.app>

Links to Code and Theses

- **GitHub Repository:** <https://github.com/IsmailAissaoui>
- **Master's Thesis (Cardiovascular Medical Digital Twin):** Available via Academic Portfolio
- **State Engineer Thesis (Gas Turbine Digital Twin):** Available via Academic Portfolio

Record of Academic Courses & Grading

Institution: Ecole Supérieure en Informatique de Sidi Bel-Abbès (ESI-SBA)

System: 0-20 scale (10 is passing; >14 is Very Good; >16 is Excellent)

- **Core Mathematics & Theory:** Calculus, Linear Algebra, Probability & Statistics, Optimization Theory.
- **Computer Science & AI:** Machine Learning, Deep Learning, Computer Vision, Signal Processing, Data Structures & Algorithms.
- *Note: Full official academic transcripts and translated degree certificates are available upon request or upon formal enrollment.*

Key Academic Projects

1. Cardiovascular Medical Digital Twin (Master's Thesis)

Contribution: Engineered "Medical_MambaTS", a linear-time state-space model for sub-millisecond bedside inference. Developed PI-DeepONet, a Physics-Informed Operator Network that explicitly maps 0D Windkessel differential equations into the neural architecture via PyTorch Autograd. Designed a "Missingness-Aware Gate" to dynamically handle sensor dropouts in ICU data.

2. Gas Turbine Digital Twin (State Engineer Thesis)

Contribution: Built predictive maintenance architectures using ST-KDFormer and Temporal Convolutional Networks to predict thermodynamic degradation and sensor drift in complex rotating machinery. Validated against real-world, high-frequency SCADA telemetry.

Academic References

Ismail Aissaoui

AI & Data Science Engineer

Reference 1

Dr. Mohammed Bedjaoui

Primary Thesis Supervisor

Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA)

Email: m.bedjaoui@esi-sba.dz

Relationship: Supervised both my Master's Thesis (Cardiovascular Medical Digital Twin) and State Engineer Thesis (Gas Turbine Digital Twin). Can comprehensively vouch for my technical research capability, engineering architecture skills, and work ethic.

Reference 2

Dr. Belkacem Khaldi

Academic Professor

Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA)

Email: b.khaldi@esi-sba.dz

Relationship: Academic professor and course instructor. Can vouch for my theoretical understanding of complex systems and overall academic excellence.